



loaders, contributing to low carbon footprint initiatives,” shares Kumar. They convert vehicles by conducting integration testing, validation, and installation in their factory. After the initial conversion, the interested party takes the vehicle to government agencies for licensing and approval.

Tsuyo also offers training and guidelines for conversion to other companies interested in this process. “We also provide ongoing support and guidance for any operational challenges or further assistance needed. Additionally, we arrange to procure other components like batteries and chargers from the market, initially integrating them into the vehicle and later allowing the companies to source these components from different suppliers,” says Kumar.

They already claim to have two manufacturing units in Greater Noida, one focusing on motor windings and the other on assembly. They are also establishing a third unit in Karnataka, South India, which will be operational by the end of December this year.

Interestingly, they are leveraging partnerships with educational institutions too. With NIT Nagpur, they



Vijay Kumar, Co-Founder, Tsuyo Manufacturing

have collaborated on power electronics projects under the mentorship of Professor Kesari, supporting students, interns, and PhD candidates while contributing to new research initiatives. They work with NIAMT Ranchi to develop new materials for motor components. Additionally, Tsuyo has a three-year MoU with IIT Delhi and is an industry partner in a government-funded project there, with a budget of 20 million rupees. They are designing a complete indigenous motor with the IIT Delhi team for this project, planning to launch it in the market once completed.

Tsuyo Manufacturing is expanding its product line to include gearboxes and axles, complementing its existing motor and controller offerings. The company has formalised the design for these products and is preparing to launch production through its investor partner’s factories. Also, they claim to have designed additional components, such as display circuits. The company seeks a manufacturing partner to produce this design as a package. Tsuyo Manufacturing plans to offload assembly and packaging to vendors who supply the components for integration with their motors.

Overall, they reported a remarkable 300% growth in the second year of operations. For the immediate future, “we remain optimistic about achieving a growth rate of 50% this year due to interim expansion and many changeovers and transformation and gearing up for the bigger leap in coming years. We focus on automating production lines and expanding our operations to enhance efficiency. We aim to partner with at least 500 car companies over the next five years,” concludes Kumar.

—Nitisha Dubey

2 STARTUP KARKHANA IO REINVENTING THE EMS WHEEL IN INDIA

EMS is a capital-intensive industry, and hence, with the strongest clashes. Then how is a startup, Karkhana IO, on its way to clock a revenue of ₹1 billion this fiscal? The founders explain!

Reinventing the wheel in the EMS industry is rarely discussed due to high capital costs involved. Thanks to its unique business model, Karkhana IO made ₹200 million last fiscal and aims for ₹1 billion this year. Sonam Motwani, Founder, and Karthik MC, CRO, Karkhana IO, told EFY’s Mukul Yudhveer Singh that their electron-

ics manufacturing services (EMS) startup aims to support medium and small industries (MSMEs) and other electronic product startups on the supply side.

“By supply, we mean the supply of manufactured electronics and solutions. In India, the demand for finished electronic goods is more than the supply of the same. This is what

we aim to solve,” says Motwani.

Karkhana IO has partnered with over 500 EMS companies. Simply put, the startup connects these companies with customers who have a demand for custom-built solutions. In the process, the responsibility of quality and delivery of solutions is completely that of the startup. However, it does not get involved in any